

CSCE 240 – Programming Assignment Six

Due: 11:59pm on Thursday, April 24

Project Purpose

Create a `TrueFalseQuestion` class as a child of the `Question` class. The class will contain one private `bool` data member for whether the answer to the question is true or false. You will include a constructor with default arguments for the question and answer, accessor and mutator functions for the private data member, and a virtual destructor and `Print` function. Read the comments in the supplied `truefalsequestion.h` header file for more detailed requirements. Initial tests for the functionality of the `TrueFalseQuestion` class have been provided in the attached `testtf.cc` source file. You are encouraged to create more rigorous tests.

Create a `ShortAnswerQuestion` class as a child of the `Question` class. The class will contain one private `string` data member for the correct answer to the question. You will include a constructor with default arguments for the question and answer, accessor and mutator functions for the private data member, and a virtual destructor and `Print` function. Read the comments in the supplied `shortanswerquestion.h` header file for more detailed requirements. Initial tests for the functionality of the `ShortAnswerQuestion` class have been provided in the attached `testsa.cc` source file. You are encouraged to create more rigorous tests.

Create a `MultipleChoiceQuestion` class as a child of the `Question` class. The class will contain three private data members: an `int` for the number of answer choices, a pointer to a `string` to hold the dynamically allocated array of answer choices, and a pointer to a `bool` to hold the dynamically allocated array of true/false values denoting whether or not each of the corresponding answer choices is correct. You will include a constructor with default arguments for the data members, a copy constructor, the accessor and mutator functions described in the header file, a virtual destructor, and a `Print` function. Read the comments in the supplied `multiplechoicequestion.h` header file for more detailed requirements. Initial tests for the functionality of the `MultipleChoiceQuestion` class have been provided in the attached `testmc1.cc` and `testmc2.cc` source files. You are encouraged to create more rigorous tests.

Your files should compile and link with the attached `testallquestions.cc` source file that reads `samplequestions.txt` of true/false, short answer, and multiple choice questions, dynamically creates objects to hold the questions, and prints them using the virtual `Print` function.

Additional Specifications

- Do not modify any of the code in `question.h`
- Add all code for the definition of the `TrueFalseQuestion` class to the attached `truefalsequestion.h` header file.
- Include all of the necessary code for the `TrueFalseQuestion` class, including the implementation of all of the public member functions, in the attached `truefalsequestion.cc` source file.
- Add all code for the definition of the `ShortAnswerQuestion` class to the attached `shortanswerquestion.h` header file.
- Include all of the necessary code for the `ShortAnswerQuestion` class, including the implementation of all of the public member functions, in the attached `shortanswerquestion.cc` source file.
- Add all code for the definition of the `MultipleChoiceQuestion` class to the attached `multiplechoicequestion.h` header file.

- Include all of the necessary code for the MultipleChoiceQuestion class, including the implementation of all of the public member functions, in the attached multiplechoicequestion.cc source file.
- You are required to use pointers as data members in the MultipleChoiceQuestion class and to manage the dynamic allocation and deallocation of memory for these data members.
- The final output statement in each Print function should end with `<< endl;`
- You will submit a zip file (only a zip file will be accepted) containing truefalsequestion.h, truefalsequestion.cc, shortanswerquestion.h, shortanswerquestion.cc, multiplechoicequestion.h and multiplechoicequestion.cc to the assignment in Blackboard.
- Source files must compile and run on a computer of the instructor's choosing in the Linux lab (see your course syllabus for additional details).
- Be sure to review the program expectations section of the course syllabus.

Initial Testing

- A makefile has been included to aid in using testtff.cc, testsa.cc, testmc1.cc, testmc2.cc, testallquestions.cc and checkit.cc to test the basic functionality of your classes. To use the makefile, your program 6 directory should include:
your files: truefalsequestion.h, truefalsequestion.cc, shortanswerquestion.h, shortanswerquestion.cc, multiplechoicequestion.h, multiplechoicequestion.cc
the provided makefile: makefile
the provided test files: testtff.cc, testsa.cc, testmc1.cc, testmc2.cc, testallquestions.cc, and checkit.cc
the provided subdirectory "output" that contains correcttff.txt, correctsa.txt, correctmc1.txt, correctmc2.txt, correctallqs.txt
- To test your TrueFalseQuestion class, type: **make testtff**
- To test your ShortAnswerQuestion class, type: **make testsa**
- To test your MultipleChoiceQuestion constructor with default arguments and Print function, type: **make testmc1**
- To test your MultipleChoiceQuestion copy constructor and SetAnswerChoices function, type: **make testmc2**
- To test the virtual Print function for all classes, type: **make testallquestions**
- Note: each of the tests above runs the executable generated by the included checkit.cc source file which compares the output created by your functions to the expected output (held in the files provided in the output folder).

Grade Breakdown

Style truefalsequestion.h: 0.1 points
 Style truefalsequestion.cc: 0.2 points
 Style shortanswerquestion.h: 0.1 points
 Style shortanswerquestion.cc: 0.2 points
 Style multiplechoicequestion.h: 0.2 points
 Style multiplechoicequestion.cc: 0.2 points
 Documentation: 1 point
 Clean compilation of truefalsequestion.cc: 0.2 points

Clean compilation of shortanswerquestion.cc: 0.2 points
Clean compilation of multiplechoicequestion.cc: 0.1 points
Clean compile/link with testtf.cc: 0.1 points
Clean compile/link with testsa.cc: 0.1 points
Clean compile/link with testmc1.cc: 0.1 points
Clean compile/link with testmc2.cc: 0.1 points
Clean compile/link with testquestions.cc: 0.1 points
Passes instructor's modified testtf.cc tests: 1 point
Passes instructor's modified testsa.cc tests: 1 point
Passes instructor's modified testmc1.cc tests: 1 point
Passes instructor's modified testmc2.cc tests: 1 point
Passes instructor's modified testallquestions.cc tests: 2 points
No memory leaks: 1 point

The penalty for late assignment submissions is 10% per day up to three days after the assignment due date. No assignment submissions will be accepted more than 3 days after the due date.